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337612(37)

B. E. (Sixth Semester) EXAMINATION, April-May, 2015

(Old Scheme)

(Branch : Mechanical)

TURBO MACHINERY

Time : Three Hours] [*Maximum Marks : 80*

[*Minimum Pass Marks : 28*

Note : (i) Attempt all the *five* questions. In each question, answer any *two* parts.

(ii) Use of Steam table and Mollier chart is permitted.

(iii) For air, assume $\gamma = 1.4$ and $C_p = 1.005$ kJ/kg K.

1. (a) What is compounding in a steam turbine ? Explain the velocity compounded impulse turbine with the help of a neat sketch. 8

(b) Steam leaves the nozzle of a single stage impulse turbine at 840 m/s. The nozzle angle is 18° and the blade angles are 29° at inlet and outlet. Due to

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friction, the blade velocity coefficient i Calculate :

(i) Blade velocity.

(ii) Steam mass flow rate in kg/h to de 300 kW power.

(c) In a simple impulse turbine, the nozzle ang 16° and the blade outlet angle is 25°. velocity of steam as it issues from nozzl 720 m/s and the blade velocity is 180 m/s. blade velocity coefficient may be taken as The steam flow rate 1 kg/s.

Find :

(i) Energy dissipated in blades due to frictio

(ii) Power developed.

(iii) Blade efficiency.

2. (a) List the various losses in steam turbine explain briefly the losses due to wetness steam.

(b) In a Parson's turbine running at 1500 r. p. m. available enthalpy drop for expansion is 63 kJ The mean diameter of the rotor is 100 Assuming that the efficiency of a stage is blade outlet angle is 20° and blade to st velocity ratio is 0.7, find the number of mov blade rows required in the turbine.

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(c) The fixed blades of stage of a Parson's turbine deliver steam at 90 m/s. The moving blade exit angle is 20°. The axial velocity of the steam is 3/4 of the blade velocity. Steam is supplied to the stage at the rate of 9000 kg/h. Calculate : 8

(i) Power developed

(ii) Blade efficiency

(iii) Enthalpy drop in the stage

3. (a) Define reheat factor, stage efficiency and internal efficiency for a multistage turbine and obtain the relationship between them. 8

(b) Explain the method of Throttle control governing of a steam turbine with the help of a neat sketch. 8

(c) A steam turbine operates at 60% load. It is governed by throttling.

During a test the following data were obtained :

Steam pressure and temperature before the governing valve = 30 bar and 400°C

Steam pressure after governing value = 10 bar

Pressure at exhaust = 0.1 bar

Steam temperature at exhaust = 55°C

Steam flow rate through turbine = 2000 kg/h

Find :

(i) Internal efficiency of the turbine.

(ii) Shaft power if the frictional losses in bearings and the power required to run the governor amounts to 8.5 kW. 8

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4. (a) Obtain the condition for optimum pressure ratio for maximum specific output in an ideal simple gas turbine cycle. 8

(b) The air enters the compressor of a gas turbine at 1 bar and 27°C and is compressed to 6 bar. The air temperature entering the turbine is 577°C. If the volume flow rate of air entering the compressor is 240 m^3/s, calculate the net power output of the cycle in MW. Also calculate its thermal efficiency. Assume that the cycle operates under ideal conditions. 8

(c) The air enters the compressor of a gas turbine at 1 bar and 30°C and is compressed to 15 bar with an isentropic efficiency of 85%. The turbine inlet temperature is 1400°C. The isentropic efficiency of the turbine is 86%. Estimate the flow of air in kg/s for a net electric power of 50 MW. Assume 2% of pressure loss in combustion chamber, 1% pressure loss in exhaust hood, mechanical efficiency of the whole assembly 98% and alternator efficiency 98%. 8

5. (a) Draw a neat sketch of various components of a centrifugal compressor and explain its working. 8

(b) A centrifugal compressor running at 16000 r.p.m. compresses the air at the rate of 10 kg/s from inlet stagnation state of 1.2 bar and 15°C. The entry to

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the impeller is radial and it leaves the impeller with a whirl velocity of 0.94 times the blade tip velocity. The stagnation pressure and temperature after compression are 4.8 bar and 202°C respectively. Calculate :

(i) Tip diameter of impeller.

(ii) Isentropic efficiency.

(iii) Power required to drive the compressor.

(c) In a 10-stage axial flow compressor, the overall compressor pressure ratio achieved is 5 : 1 with an overall isentropic efficiency of 87%. The temperature and pressure at inlet is 15°C and 1 bar. The work is divided equally between the stages. A 50% reaction is used with a blade speed of 210 m/s and a constant axial velocity of 170 m/s. Calculate the blade angles at inlet and outlet with respect to axial direction. 8

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